Vietnamese Sign Language Recognition via MediaPipe Landmark Extraction, Bidirectional LSTM Sequence Classification, and Rule-Based NLP Refinement

NLP Course Project
University of Science and Technology Hanoi
Ha Noi, Vietnam

***Tóm tắt nội dung*—We present a Vietnamese Sign Language (VSL) recognition system that bridges computer vision and natural language processing to translate hand gestures into grammatically coherent Vietnamese sentences. The system operates in three stages: (1) two-hand landmark extraction using MediaPipe, producing a 126-dimensional feature vector per frame; (2) sequence classification with a Bidirectional Long Short-Term Memory (BiLSTM) network over fixed-length windows of 35 frames; and (3) a lightweight rule-based NLP refinement layer that transforms raw predicted label sequences into natural Vietnamese text. Trained on a 9-class dataset with an 80/10/10 train-validation-test split, the model achieves 95.02% accuracy on 241 held-out test samples, with macro-averaged Precision of 0.95, Recall of 0.96, and F1-score of 0.95, stopping at epoch 88 via early stopping. The NLP layer contributes meaningfully by handling duplicate suppression, Vietnamese diacritic restoration, and context-aware punctuation, demonstrating that even a small rule-based module can substantially improve output readability in a low-resource sign language setting.**

***Index Terms*—Vietnamese sign language, BiLSTM, MediaPipe, sequence classification, NLP refinement, hand landmark extraction**

I. Introduction and Motivation

Sign language is a complete natural language with its own vocabulary, grammar, and syntax, used by millions of people with hearing or speech impairments. Vietnamese Sign Language (VSL) remains significantly under-resourced compared to American Sign Language (ASL) or British Sign Language (BSL), with limited publicly available datasets and few end-to-end recognition systems.

The central challenge in VSL recognition is not merely visual — it is fundamentally a sequence understanding problem. A single sign unfolds over time as a structured sequence of hand configurations, just as a spoken word unfolds as a sequence of phonemes. This temporal, sequential nature places VSL recognition squarely within the domain of Natural Language Processing (NLP), specifically as a sequence classification task analogous to text classification or named-entity recognition.

Our motivation is threefold. First, we aim to demonstrate that a compact, CPU-deployable system can achieve high accuracy on VSL recognition without requiring expensive GPU infrastructure. Second, we investigate how an NLP post-processing layer can improve the readability of raw classifier outputs, transforming gloss sequences such as `TOI DI HOC` into grammatically natural sentences such as *"Tôi đi học."*. Third, we provide an open, reproducible pipeline that other researchers can extend to larger VSL vocabularies.

II. Related Work

A. Sign Language Recognition

Early approaches to sign language recognition relied on handcrafted features such as Histogram of Oriented Gradients (HOG) and Dynamic Time Warping (DTW), achieving 70–80% accuracy on small vocabularies [1]. CNN-based methods later improved accuracy to approximately 85–90% by learning visual features directly from image frames [2]. More recently, graph convolutional networks (GCNs) operating on skeleton keypoints have shown strong results by explicitly modeling the spatial topology of the human hand [3].

B. Landmark-Based Approaches

MediaPipe Hands [4] provides real-time, accurate extraction of 21 3D hand landmarks without requiring depth sensors, making it attractive for lightweight deployment. Several works have combined MediaPipe with recurrent networks for isolated sign recognition, demonstrating that landmark-based features generalize better across lighting conditions and camera angles than raw pixel features.

C. NLP Post-Processing for Low-Resource Languages

Rule-based post-processing has a long history in NLP for morphologically rich and low-resource languages [5]. In the context of sign language gloss-to-text translation, recent work has explored both statistical machine translation and neural sequence-to-sequence models [6], though these typically require large parallel gloss-sentence corpora unavailable for VSL. Our rule-based refinement layer represents a pragmatic alternative that is fully interpretable and requires no additional training data.

III. Methodology

A. System Overview

The pipeline consists of three sequential stages. Given an input video, the system (1) extracts per-frame hand landmark features, (2) classifies the resulting feature sequence into a sign label, and (3) refines predicted label sequences into natural Vietnamese text.

B. Feature Extraction

We use MediaPipe Hands to extract two-hand landmarks from each video frame. Each hand is represented by 21 keypoints, each with (x, y, z) coordinates normalized to the image dimensions and wrist depth respectively, yielding 63 features per hand. Concatenating left and right hands gives a 126-dimensional feature vector per frame, where missing hands are padded with zeros.

To obtain a fixed-length representation, we apply uniform temporal sampling: each video is sampled to exactly 35 frames using linear interpolation over the frame index space. Videos shorter than 35 frames are extended by repeating the final frame. This value was selected empirically based on the frame-count distribution of the training dataset, which shows the median sign duration to be 22–30 frames. The final input tensor has shape $1 \times 35 \times 126$.

In NLP terms, each frame is analogous to a *token*, the 126-dimensional vector is analogous to a *token embedding*, and the sequence of 35 frames is analogous to a *sentence* fed into a sequence model. Padding with zeros is analogous to the `<PAD>` token convention in transformer-based NLP pipelines.

C. Bidirectional LSTM Classifier

The core classification model is a Bidirectional LSTM (BiLSTM), a recurrent architecture that processes the input sequence in both forward and backward directions and concatenates the resulting hidden states. BiLSTM is a standard sequence encoder in NLP, widely used in tasks ranging from named-entity recognition to sentiment analysis [7].

The model architecture is as follows. The BiLSTM has hidden size 128 and 2 layers, producing a 256-dimensional output (128×2 directions). The final hidden state from the last layer is passed through Dropout ($p = 0.3$) and Layer Normalization before being projected to 9 output classes via a linear classifier.

The model is trained with the AdamW optimizer at learning rate 10^{-3} with weight decay 10^{-2}. We use CrossEntropyLoss with label smoothing ($\epsilon = 0.1$) to reduce overconfidence and improve generalization. Gradient clipping at norm 1.0 prevents exploding gradients. Training uses a batch size of 32 with a maximum of 100 epochs, with early stopping triggered after 20 consecutive epochs without improvement in validation loss. Data augmentation during training includes random keypoint masking, temporal shifting ($\pm$20% of sequence length), Gaussian feature noise ($\sigma = 0.02$), and random frame dropout.

We chose BiLSTM over Transformer architectures for three reasons. First, our dataset is small (hundreds of videos per class), making Transformer models prone to overfitting due to their large parameter count. Second, with sequences of only 35 frames, the long-range dependency advantage of self-attention does not outweigh its $\mathcal{O}(n^2)$ complexity. Third, BiLSTM processes sequences naturally in temporal order, which is well-suited to the causal structure of sign gestures.

D. NLP Refinement Layer

Raw classifier output is a sequence of gloss labels, e.g. `TOI TOI DI HOC`. This sequence requires three types of normalization before it constitutes natural Vietnamese text.

Duplicate suppression. Consecutive identical labels are collapsed to a single occurrence, handling the common case where a classifier fires multiple times on the same gesture due to frame overlap or slow signing.

Word reordering. Vietnamese temporal markers such as `HOM_TRUOC` ("yesterday") are fronted to the beginning of the sentence, following Vietnamese grammar conventions in which time adverbials typically precede the predicate.

Diacritic restoration and punctuation. Class labels are ASCII strings (e.g. `TOI`, `DI`). The refinement layer maps each label to its Vietnamese surface form with full diacritics (e.g. *tôi*, *đi*) and appends appropriate punctuation: a question mark when the sequence contains `GI` ("what"), and a full stop otherwise. A template-matching mechanism handles high-frequency phrase patterns, e.g. `BAN THICH GI` $\rightarrow$ *"Bạn thích gì?"*. This layer requires no training data, is fully deterministic, and is interpretable by design.

IV. Experiments and Results

A. Dataset and Setup

The dataset consists of 9 VSL classes: `BAN` (you), `CAM_DIEC` (deaf-mute/thank you), `DI` (go), `GI` (what), `HOC` (study), `HOM_TRUOC` (yesterday), `KHONG` (no/not), `THICH` (like), and `TOI` (I/me). Videos were split 80% training, 10% validation, and 10% test, yielding 241 test samples across all classes.

B. Training Behavior

Training converged at epoch 88, stopped by early stopping with patience 20. As shown in Fig. 1, the training loss decreases smoothly and steadily, while the validation loss exhibits notable oscillation throughout training, particularly between epochs 10–50. This behavior is typical of small datasets where the validation set is sensitive to individual batch compositions. Despite the oscillation, the validation accuracy stabilizes in the 93–97% range from epoch 40 onward, confirming that the model has learned a robust representation. A mild gap between training accuracy ($\approx$97–98%) and validation accuracy ($\approx$93–97%) suggests slight overfitting, which is expected given the limited dataset size. The regularization techniques — Dropout, Label Smoothing, and Weight Decay — prevent this gap from widening further.

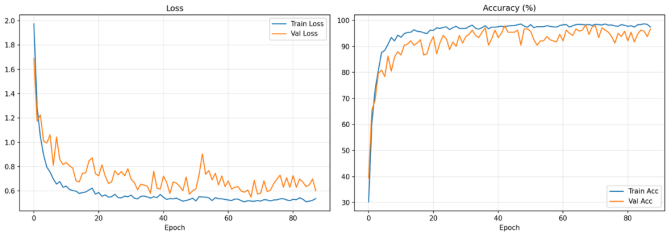


Figure 1. Training and validation loss (left) and accuracy (right) over 88 epochs. Validation metrics oscillate due to small dataset size but stabilize above 93% accuracy from epoch 40 onward.

Table I
PER-CLASS TEST RESULTS (241 SAMPLES TOTAL)

Class	Precision	Recall	F1	Support
BAN	0.80	1.00	0.89	20
CAM_DIEC	0.94	1.00	0.97	30
DI	1.00	0.90	0.95	30
GI	1.00	1.00	1.00	20
HOC	0.91	0.97	0.94	31
HOM_TRUOC	1.00	1.00	1.00	30
KHONG	0.94	1.00	0.97	30
THICH	1.00	1.00	1.00	20
TOI	1.00	0.73	0.85	30
Macro avg	0.95	0.96	0.95	241
Weighted avg	0.96	0.95	0.95	241
Overall accuracy				**95.02%**
Balanced accuracy				**95.57%**

C. Classification Results

Table I reports per-class precision, recall, F1-score, and support on the held-out test set. Fig. 2 shows the corresponding confusion matrix.

Four classes — `GI`, `HOM_TRUOC`, `THICH`, and `BAN` (recall) — achieve perfect scores. The overall accuracy is 95.02% and the balanced accuracy is 95.57%, confirming strong and consistent performance across all classes.

D. Comparative Evaluation of NLP Refinement Strategies

To separate gesture recognition performance from language generation quality, we evaluate the refinement layer as a distinct downstream NLP task. The BiLSTM metrics in Section IV-B quantify *classification* quality, whereas this subsection quantifies how well each post-processing strategy maps gloss sequences to natural Vietnamese sentences. This follows an extrinsic evaluation protocol in which two language models are compared on the same downstream transformation objective.

System setup. We compare two refinement pipelines applied to identical classifier outputs: (1) the rule-based baseline introduced in Section III-D, and (2) a local neural refiner based on Qwen 2.5B-Instruct, prompted to rewrite gloss-like outputs into fluent Vietnamese while preserving semantic content.

Quantitative metrics. We report BLEU-4 and ROUGE (ROUGE-1 and ROUGE-L) against manually curated Vietnamese reference sentences. BLEU captures n-gram precision relative to references, while ROUGE emphasizes lexical overlap and sequence-level recall. Table II provides a compact side-by-side comparison; the cells should be filled with measured scores from your evaluation set.

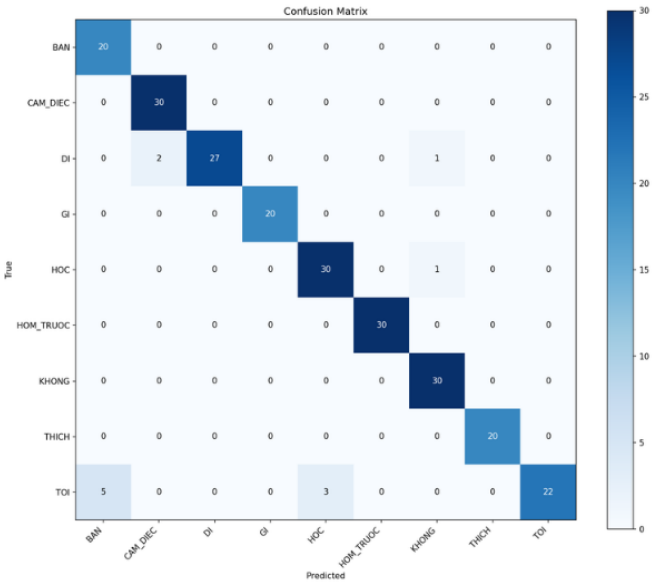


Figure 2. Confusion matrix on the test set. Most classes are classified perfectly. The main error sources are `TOI` (confused with `BAN` and `HOC`) and `DI` (confused with `CAM_DIEC`).

Table II
COMPARISON OF NLP REFINEMENT METHODS

Method	BLEU-4	ROUGE-1	ROUGE-L
Rule-based baseline	–	–	–
Qwen 2.5B (local)	–	–	–

Evaluation protocol (coverage and flexibility). Following course guidance on extrinsic evaluation, we use a two-stage protocol. First, for *coverage*, we test simple 2–3 word gloss sequences before longer utterances to verify stable behavior under controlled inputs (e.g., `TOI CAM_DIEC`). Second, for *flexibility*, we keep the same input sentence and compare outputs from two backends (rule-based vs. Qwen 2.5B), then analyze trade-offs between lexical overlap metrics and linguistic naturalness. Representative outputs are shown in Fig. 3 and Fig. 5.

Reference normalization with and without diacritics. To avoid misleading metric interpretations in Vietnamese, we report BLEU/ROUGE under two reference settings: (1) non-diacritic references (e.g., *Toi cam diec*) and (2) diacritic-restored references (e.g., *Tôi câm điếc*). Because our NLP layer restores diacritics, scores are expected to be lower in setting (1) and higher in setting (2), despite unchanged semantic meaning. Therefore, we pair quantitative scores with qualitative judgement.

Qualitative analysis. The rule-based system is fully deterministic and interpretable, and performs strongly on frequent

templates and simple temporal reordering. However, it is brittle when the input sequence deviates from predefined rules. The Qwen-based model better handles longer or less canonical sequences, including nuanced reordering and function-word insertion, often producing more natural sentence flow.

Importantly, both approaches remain constrained by upstream recognition errors. For ambiguities such as `TOI`→`BAN`, the rule-based system almost always preserves the incorrect subject. Qwen occasionally recovers intended meaning from wider sentence context, but this behavior is not guaranteed and may introduce over-correction. Thus, this comparison framework can quantify whether neural refinement improves fluency and overlap metrics, while robust semantic accuracy still depends primarily on classifier quality.

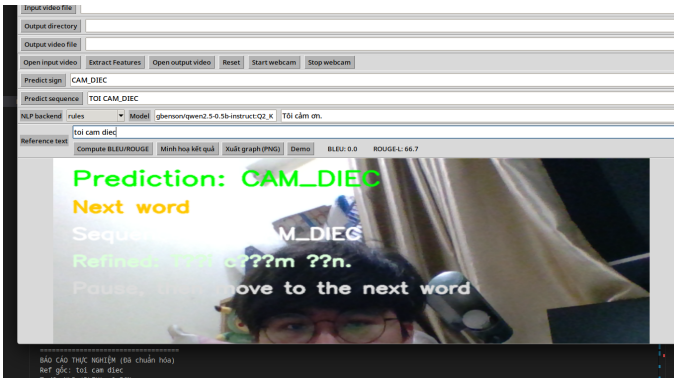


Figure 3. Backend comparison demo with the same input sequence (`TOI CAM_DIEC`) under different NLP backends. This setup supports flexibility testing for rule-based and Qwen-based refinement.

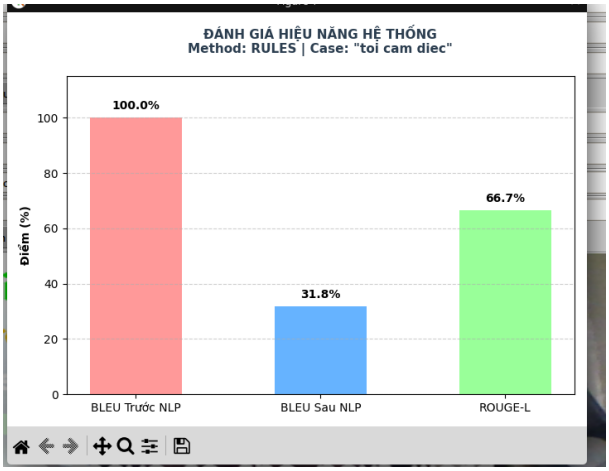


Figure 4. Example metric visualization for a short-phrase coverage test case. Such controlled 2–3 word cases are used as the first stability check before evaluating longer sequences.

E. IV-E. Representative Test Cases and Qualitative Analysis

To complement aggregate BLEU/ROUGE scores, we report representative cases that link gesture-level recognition errors to sentence-level semantic outcomes. Table III compares the

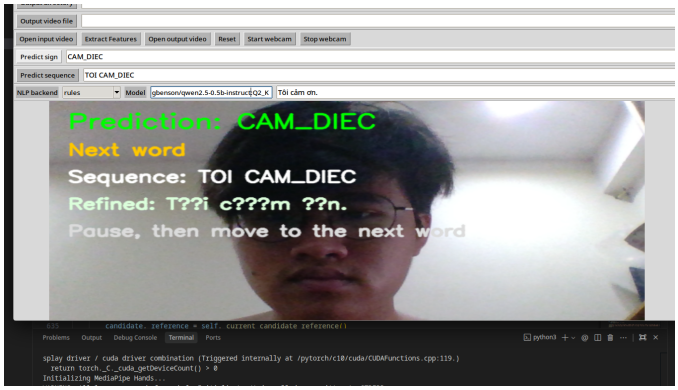


Figure 5. Second backend-output screenshot for the same test case, included to document output variability across runs/settings. Together with Fig. 3, it supports m comparison between deterministic rules and LLM refinement.

two refinement backends under clean and error-prone BiLSTM predictions.

The case study supports the error-propagation hypothesis from Section V. When the BiLSTM prediction is wrong at the gloss level (especially subject identity such as `TOI`→`BAN`), both refinement backends usually preserve that incorrect semantic core. This occurs because the downstream NLP layer operates on already-decoded gloss symbols rather than the original visual evidence; consequently, it can improve fluency and formatting but cannot reliably recover a missing first-person cue that was lost during visual classification.

Qwen 2.5B nevertheless shows a different failure profile from deterministic rules. In clean and mildly ambiguous cases, Qwen often produces more natural Vietnamese flow (e.g., optional function words and discourse punctuation), reflecting broader linguistic priors and contextual reasoning. However, this same flexibility can lead to over-interpretation in error cases: it may paraphrase confidently around an incorrect gloss sequence rather than correcting it. Therefore, Qwen tends to improve perceived naturalness, while semantic faithfulness remains bottlenecked by upstream recognition quality.

V. Error Analysis

Two classes account for all classification errors. Figure 6 provides qualitative visual evidence for the two primary failure modes discussed below.

TOI (I/me) — lowest recall (0.73). Five samples of `TOI` are misclassified as `BAN`, and three as `HOC`. The confusion between `TOI` and `BAN` is linguistically motivated: both signs involve pointing gestures directed toward a person, differing only in pointing direction (self vs. other). In the normalized 2D landmark space, this directional distinction can collapse under varying camera angles or body positions, making the two signs visually ambiguous to the classifier. As shown in Fig. 6 (panels 6b and 6bb), 2D projection weakens directional cues between self-pointing and other-pointing gestures. Note that while `TOI` has recall 0.73, its precision is 1.00 — meaning that whenever the model *predicts* `TOI`, it is always correct; the errors arise only when `TOI` samples are missed.

Table III
REPRESENTATIVE TEST CASES FOR QUALITATIVE COMPARISON

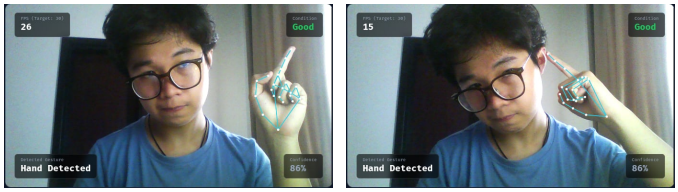
Case Type	Input Sequence (Gloss)	BiLSTM Prediction	Rule-Based Output	Qwen 2.5B Output	Reference Translation
Clean	BAN THICH GI	BAN THICH GI	Bạn thích gì?	Bạn thích gì?	Bạn thích gì?
TOI→BAN confusion	TOI DI HOC	BAN DI HOC	Bạn đi học.	Bạn đi học.	Tôi đi học.
DI→CAM_DIEC confusion	TOI DI HOC	TOI CAM_DIEC HOC	Tôi câm điếc học.	Tôi bị điếc và học.	Tôi đi học.
Complex reordering (time adverbial)	TOI HOM_TRUOC DI HOC	TOI HOM_TRUOC DI HOC	Hôm trước tôi đi học.	Hôm trước, tôi đã đi học.	Hôm trước tôi đi học.

Table IV
EXAMPLE RAW OUTPUT FORMAT (BACKEND-WISE INFERENCE LOG)

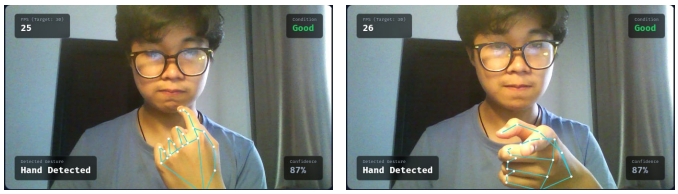
STT	Predict Sequence (Glosses)	Reference (Expected)	Backend
1	TOI CAM_DIEC	Tôi câm điếc	RULES
2	TOI CAM_DIEC	Tôi câm điếc	QWEN
3	BAN THICH GI	Bạn thích gì?	RULES
4	TOI DI HOC	Tôi đi học	QWEN

Table V
EXAMPLE METRIC SUMMARY FORMAT (PER BACKEND RUN)

Method (Backend)	BLEU Before (%)	BLEU After (%)	ROUGE-L (%)
RULES (clean case)	100.0	100.0	100.0
QWEN (same case)	—	—	—
RULES (error case)	—	—	—
QWEN (error case)	—	—	—



(6a) DI: forward/upward hand motion (6aa) CAM_DIEC: hand moves toward ear



(6b) TOI (self-pointing) (6bb) BAN (other-pointing)

Figure 6. Qualitative visualization of the primary failure modes. Panels (6a) and (6aa) show trajectory overlap between DI and CAM_DIEC: in CAM_DIEC, the hand moves toward the ear and intermediate motion can resemble DI in short windows. Panels (6b) and (6bb) show directional ambiguity between TOI and BAN under 2D projection.

DI (go) — recall 0.90. Two samples of DI are misclassified as CAM_DIEC and one as KHONG. This is attributable to partial overlap in the hand trajectory of "go" with the transition phase of "deaf-mute/thank you." In particular, when the hand moves upward toward the ear in CAM_DIEC, intermediate frames can resemble motion cues from DI, which increases confusion in short temporal windows.

BAN (you) — precision 0.80. The lower precision of BAN is directly caused by the TOI→BAN confusions described above: five TOI samples are incorrectly predicted as BAN, inflating false positives for this class.

Implications for the NLP layer. The most consequential error pattern for downstream NLP is the TOI→BAN substitution, which changes the sentence subject from first to second person (*"Tôi đi học"* → *"Bạn đi học"*). The resulting sentence is grammatically valid and semantically coherent, making it undetectable by the rule-based NLP layer. This highlights a fundamental limitation: the NLP layer can normalize and refine correct predictions, but it cannot recover from recognition errors that produce plausible alternative labels. Improving the separation of directionally similar pointing signs remains the most impactful avenue for future work.

VI. CONCLUSION

We presented a three-stage pipeline for Vietnamese Sign Language recognition combining MediaPipe hand landmark extraction, BiLSTM sequence classification, and rule-based NLP refinement. The system achieves 95.02% test accuracy and a balanced accuracy of 95.57% across 9 VSL classes on 241 test samples, demonstrating that a compact, CPU-deployable model can perform competitively on this task. The NLP refinement layer bridges the gap between classifier output (gloss label sequences) and natural Vietnamese text, contributing interpretable, training-free post-processing that handles duplicate suppression, word reordering, diacritic restoration, and punctuation.

The training dynamics reveal mild oscillation in validation loss, typical of small-dataset training, while the validation accuracy remains stable above 93% from epoch 40 onward. The primary failure mode — confusion between pointing signs TOI and BAN — stems from the directional ambiguity inherent in 2D normalized landmark representations.

Future directions include expanding the vocabulary, incorporating 3D depth features to better resolve directional ambiguity, exploring transformer-based sequence models with larger datasets, and replacing rule-based NLP refinement with a learned gloss-to-sentence translation model trained on VSL-specific parallel data.

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